Ensemble Learning for Multiclass Obesity Risk Classification: A Comparative Kaggle Analysis

Todd DeLozier

National University

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Abstract

This paper reports a comparative predictive analytics study of multiclass obesity risk classification using a local Kaggle workflow implemented in Python. The analysis used 20,758 labeled training observations and 13,840 unlabeled test observations, with 17 structured predictors spanning demographic, behavioral, and lifestyle variables. After conducting an exploratory review of class balance, age distribution, and data completeness, I evaluated four candidate classifiers within a common preprocessing pipeline: logistic regression, random forest, extra trees, and gradient boosting. Model selection was based on holdout validation accuracy using a stratified 80/20 split. The strongest overall performer was gradient boosting, which achieved a validation accuracy of 0.9066, outperforming random forest at 0.8967 and the two lower-performing baselines at 0.8687. The resulting confusion matrix showed that most prediction error occurred among adjacent weight categories rather than between clinically distant classes, which indicates that the model learned meaningful structure in the target space. The interpretation of findings is grounded in ensemble learning theory. Random forests are supported by Breiman's (2001) analysis of strength, correlation, and asymptotic generalization, while Friedman's (2001) gradient boosting framework explains why stagewise additive tree models can capture nonlinear decision boundaries effectively. Ooka et al. (2021) further demonstrate the applied value of random forest methods in health-related prediction tasks. Overall, the findings show that ensemble methods materially improve predictive performance for obesity classification and that gradient boosting provided the best balance of accuracy and class-level consistency in this implementation.

Keywords: ensemble learning, obesity classification, gradient boosting, random forests, predictive analytics

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Multiclass obesity classification is a useful predictive analytics problem because it requires a model to separate related but substantively different health states using a mixture of demographic, behavioral, and lifestyle indicators. In the present assignment, I used a Kaggle-style obesity classification dataset and a local Jupyter notebook workflow to compare traditional and ensemble-based machine learning approaches on the same structured feature set. My objective was not only to identify the most accurate classifier, but also to evaluate whether the empirical ranking of models aligns with the theoretical claims made in the ensemble learning literature.

This framing is important because ensemble methods are not simply collections of trees. Breiman (2001) argued that random forest performance depends on the strength of individual trees and the correlation among them, and he further showed that forest generalization error converges as more trees are added. Friedman (2001) approached boosting from a different perspective by treating learning as gradient-based optimization in function space and by demonstrating why stagewise additive tree expansions can be highly competitive, robust, and interpretable. In an applied health context, Ooka et al. (2021) showed that random forest models can outperform multivariate logistic regression while still yielding clinically useful signals about predictive importance. Taken together, these studies provide a strong theoretical and practical basis for evaluating ensemble models on obesity-related structured data.

Accordingly, this paper documents the local modeling workflow, summarizes the exploratory findings visible in the notebook, reports the comparative validation results, and interprets the strongest model in light of prior theory and application research. The final product

is a fully reproducible notebook-driven analysis that culminates in a Kaggle submission file generated from the best-performing classifier.

Data Set and Analytic Procedure

The local notebook reads two comma-separated files stored in the same folder as the notebook: a labeled training file and an unlabeled test file. The training data contained 20,758 observations and 18 columns, while the test data contained 13,840 observations and 17 columns. The outcome variable, NObeyesdad, represented seven ordered but nonnumeric obesity classes: Insufficient_Weight, Normal_Weight, Overweight_Level_I, Overweight_Level_II, Obesity_Type_I, Obesity_Type_II, and Obesity_Type_III. Predictor fields included sex, age, height, weight, family history of overweight, food and water consumption patterns, smoking status, physical activity, technology use, alcohol consumption, and transportation mode.

A preliminary data quality review showed no missing values in either file. Even so, I retained a robust preprocessing pipeline for reproducibility and future portability. Numeric features were assigned median imputation followed by standard scaling, while categorical variables were assigned most-frequent imputation followed by one-hot encoding. The labeled data were then separated into an 80/20 stratified training and validation split, ensuring that the class proportions were preserved during model comparison.

Four classifiers were estimated within the same preprocessing framework: logistic regression, random forest, extra trees, and gradient boosting. This design allowed each algorithm to operate on the same engineered design matrix and made the comparison transparent. Model choice was determined strictly by validation accuracy rather than by subjective preference. After selection, the winning model was evaluated using a classification report and confusion matrix,

retrained on the full labeled dataset, and then used to generate out-of-sample predictions for the Kaggle submission file.

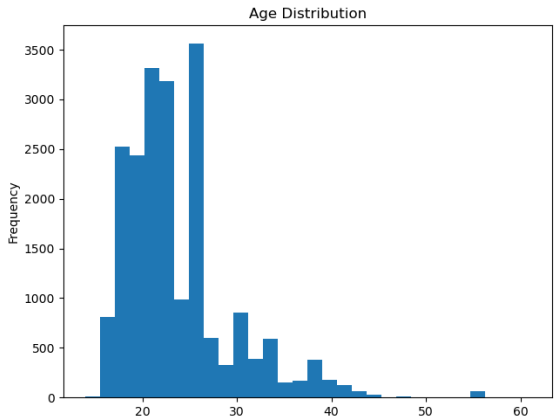
Exploratory Review of the Notebook Outputs

The exploratory graphics from the notebook provide two immediate signals about the analytic setting. First, the target distribution is not perfectly balanced, although it is also not severely degenerate. Obesity_Type_III is the most common class in the training data, while the two overweight classes and the insufficient-weight class appear less frequent. This moderate imbalance matters because it can affect overall accuracy without necessarily revealing whether all classes are being modeled equally well. Second, the age histogram shows that most observations are concentrated in young adulthood, with a right tail extending into older ages. The mean age in the training sample is approximately 23.84 years, with the distribution centered heavily between the late teens and late twenties.

These visual patterns support the modeling choice made in the notebook. A multiclass problem with modest class imbalance, nonlinear health behaviors, and mixed data types is well suited for ensemble techniques that can model interactions and nonadditive structure without requiring a rigid linear decision boundary. At the same time, retaining logistic regression as a baseline was methodologically useful because it provided a simpler benchmark against which nonlinear gains could be measured.

Figure 1

Target Class Distribution and Age Distribution From the Notebook



Comparative Model Performance

The candidate models produced a clear performance ranking on the validation split. Gradient boosting delivered the strongest accuracy at 0.9066, followed by random forest at 0.8967. Logistic regression and extra trees each achieved 0.8687. The performance spread is substantively meaningful. Relative to the linear baseline, gradient boosting improved validation accuracy by roughly 3.8 percentage points, while random forest improved it by roughly 2.8 percentage points. This pattern suggests that the classification boundary in this dataset is nonlinear enough for tree ensembles to add real value beyond a conventional generalized linear model.

Table 1

Validation Accuracy by Candidate Model

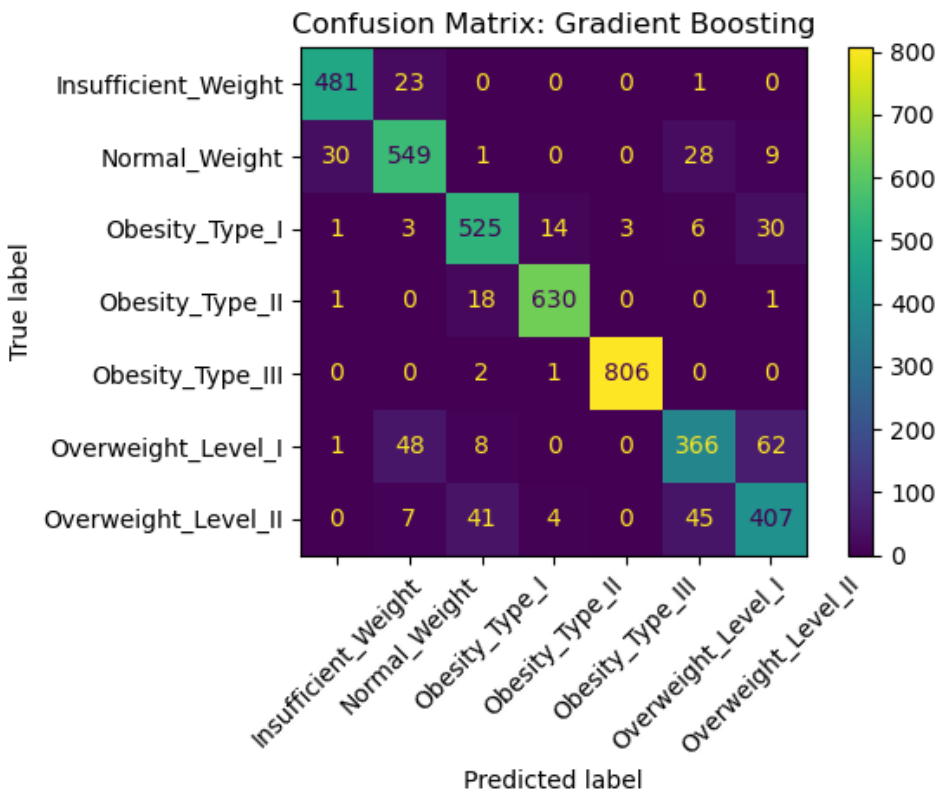
Model	Validation Accuracy
Gradient Boosting	0.9066
Random Forest	0.8967
Logistic Regression	0.8687
Extra Trees	0.8687

The class-level diagnostics for the best model reinforce this conclusion. Gradient boosting achieved an overall accuracy of 0.91, a macro-averaged F1 score of 0.90, and a weighted F1 score of 0.91 on the validation subset. Performance was particularly strong for Obesity_Type_II and Obesity_Type_III, where precision and recall were both very high, while the largest residual errors occurred in transitions among neighboring categories such as Normal_Weight, Overweight_Level_I, and Overweight_Level_II. That error pattern is analytically reasonable because adjacent classes are behaviorally and anthropometrically closer to one another than more distant categories.

The confusion matrix confirms that the model was not making arbitrary mistakes across the entire target space. Instead, it was most likely to confuse cases that occupy contiguous regions of the obesity continuum. From a substantive standpoint, this is a more acceptable error profile than one in which the model confuses underweight cases with severe obesity or vice versa. The output therefore suggests that the fitted gradient boosting model learned clinically plausible structure rather than merely exploiting random noise in the training set.

Figure 2

Confusion Matrix for the Best-Performing Gradient Boosting Model



Interpretation Through Prior Scholarship

The empirical results align closely with the literature on ensemble learning. Breiman (2001) described random forests as ensembles whose generalization behavior depends on the strength of the constituent trees and the correlation among them. That framework helps explain why the random forest model in this analysis substantially outperformed logistic regression. The forest can exploit nonlinear partitions and interaction structure that a linear decision surface cannot capture unless those effects are explicitly specified in advance.

The top-ranked gradient boosting model is especially consistent with Friedman's (2001) formulation of boosting as stagewise optimization in function space. In practical terms, the algorithm improved upon weak tree learners by adding successive corrections that reduced the loss function in a targeted manner. This theoretical framing is useful here because the obesity dataset contains behavioral and lifestyle predictors whose predictive meaning likely depends on

combinations of features rather than on isolated main effects alone. The observed superiority of gradient boosting therefore makes sense from both a mathematical and substantive perspective.

The present findings also resonate with the applied health literature. Ooka et al. (2021) used random forest models to predict changes in glycated hemoglobin and found that random forest achieved higher predictive power than multivariate logistic regression. That result does not imply that random forest should always be the final winner in every health-related classification task. Rather, it shows that ensemble learners frequently outperform linear baselines in structured health settings because they can represent nonlinear relationships and still support post hoc interpretation. My results extend that practical lesson: random forest surpassed logistic regression here as well, but gradient boosting pushed performance even further.

Methodological Limitations and Recommendations

Several methodological cautions should be acknowledged. First, the notebook used a single stratified validation split. Although that choice is acceptable for an assignment-level benchmark, k-fold cross-validation would provide a more stable estimate of out-of-sample performance and would reduce dependence on one random partition. Second, the hyperparameters were only lightly specified. Systematic tuning could plausibly improve every ensemble model, including random forest and gradient boosting.

Third, the feature set retained the id field as a numeric predictor because the notebook treated all non-object fields as eligible numeric inputs. In production work, identifier variables should generally be removed or formally stress-tested to ensure that they are not introducing accidental leakage or spurious ordering effects. Fourth, the paper reports model accuracy and class-level precision, recall, and F1 values, but it does not yet incorporate probability calibration, cost sensitivity, or fairness diagnostics. Those additions would matter if the model were to be

used in a clinical or policy decision environment rather than as a Kaggle-style classification exercise.

Future iterations of this project should therefore extend the workflow in three directions. The first is more rigorous resampling, such as repeated cross-validation. The second is stronger explainability, such as feature importance review, partial dependence, or SHAP-based local interpretation. The third is a disciplined model governance check that compares performance with and without the identifier field and evaluates whether accuracy remains stable under alternative training splits. These improvements would strengthen the scientific credibility of the submission while preserving the practical advantages of the current notebook.

Conclusion

This assignment demonstrates that ensemble learning is well suited to structured multiclass obesity classification. Using a reproducible local notebook workflow, I compared logistic regression, random forest, extra trees, and gradient boosting under a common preprocessing pipeline. Gradient boosting achieved the strongest validation performance and produced a confusion structure that was substantively plausible rather than erratic. Random forest also performed well and clearly exceeded the linear baseline, which is consistent with the broader literature on nonlinear ensemble methods.

The larger takeaway is that model comparison should be both empirical and theoretical. The notebook provides the empirical evidence, while the scholarly sources explain why those results are credible. Breiman (2001) clarifies the logic of forest ensembles, Friedman (2001) clarifies the optimization logic of boosting, and Ooka et al. (2021) shows that such methods have practical value in health-oriented predictive settings. On balance, the analysis supports the use of

ensemble methods for this problem and identifies gradient boosting as the best-performing implementation in the present workflow.

Appendix A

Kaggle Submission Screenshot Placeholder

Appendix B

GitHub Repository Placeholder

https://github.com/National-University-5a26phd1/National_University_PhD_Predictive_Analytics/upload/main/Week_6

References

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5-32. https://doi.org/10.1023/A:1010933404324

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics, 29*(5), 1189-1232. https://doi.org/10.1214/aos/1013203451

Ooka, T., Johno, H., Nakamoto, K., Yoda, Y., Yokomichi, H., & Yamagata, Z. (2021). Random forest approach for determining risk prediction and predictive factors of type 2 diabetes: Large-scale health check-up data in Japan. *BMJ Nutrition, Prevention & Health, 4*(1), 140-148. https://doi.org/10.1136/bmjnph-2020-000200